

JOB SEQUENCING (GREEDY METHOD)

Job Sequencing (Greedy Method)

Let there be 14 jobs with the following profits:

22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27 and the corresponding deadlines:

3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 9, 14, 1. Find the optimal job sequence that maximizes the total profit using the greedy method.

ALGORITHM:

Job Sequencing (greedy method)

Q) Let there be 14 jobs with the profit
22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27
with deadlines - 3, 3, 8, 6, 7, 5, 10, 4, 6, 12,
13, 9, 14, 1

Sol:-

No. of jobs (N) = 14

Profits corresponding to Jobs J_1 to J_{14} &
 P_1 to P_{14} are given in question

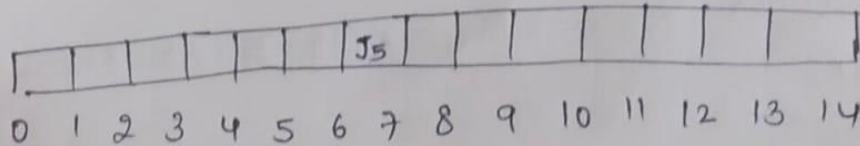
Step-1:-
Arrange the jobs in descending order based
profit and write corresponding deadlines.

30	29	28	27	27	26	25	24	22	21	19	19	14	11
7	8	6	10	2	12	4	6	3	5	3	14	13	1
J_5	J_3	J_4	J_7	J_{12}	J_{10}	J_8	J_9	J_1	J_6	J_2	J_{13}	J_{11}	J_{14}

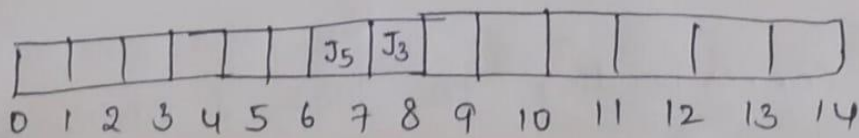
Step-2:-

Create Slots & Assign Jobs

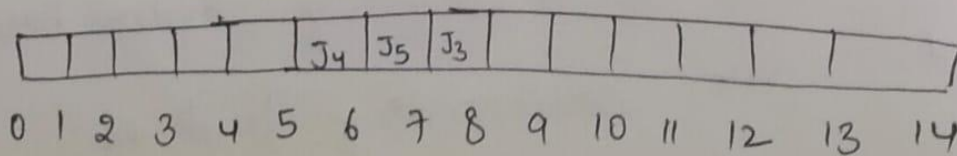
1) J_5 , $P_5 = 30$; $P_5 = 7$



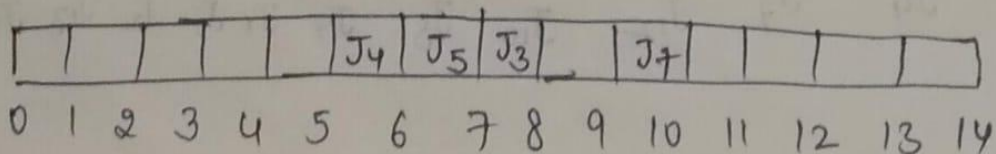
2) J_3 ; $P_5 = 29$, $D = 8$



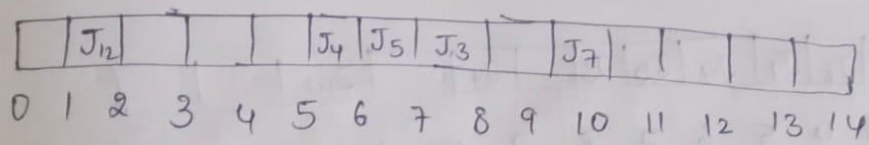
3) J_4 , $P_4 = 28$, $p_4 = 6$



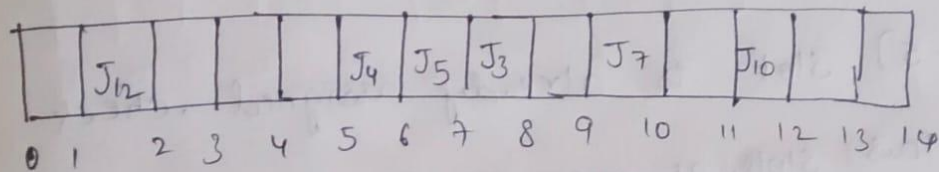
4) J_7 , $P_7 = 27$, $P_7 = 10$



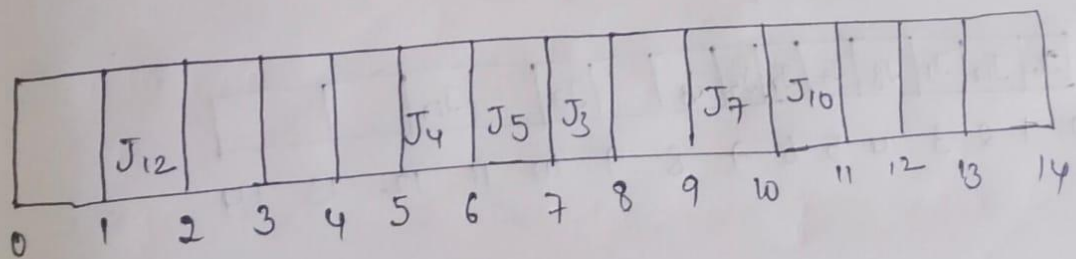
5) J_{12} , $P_{12} = 27$, $D_{12} = 2$



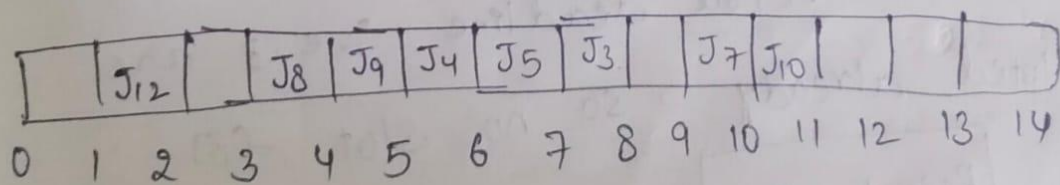
6) J_{10} , $P_{10} = 26$, $D_{10} = 12$



7) J_8 , $P_8 = 25$, $P_8 = 4$



8) J_9 , $P_9 = 24$, $P_9 = 6$



9) J_1 , $P_1 = 22$, $D_1 = 3$

	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}			
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

10) J_6 , $P_6 = 21$, $D_6 = 5$

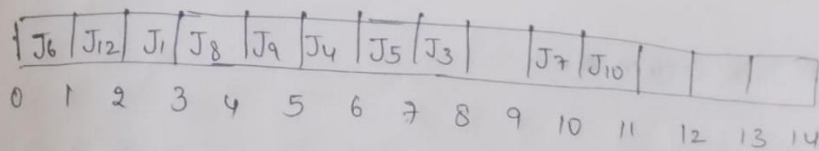
[4-5] slot is already assigned, check previous slots, as only [0,1] is free a lot of with ~~J_6~~ J_6 .

J_6	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}			
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

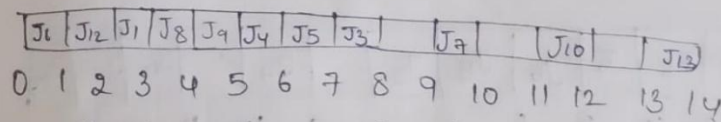
11) J_2 , $P_2 = 19$, $D_2 = 3$

All slots before deadline, i.e., 3 are allotted already. So no slot for

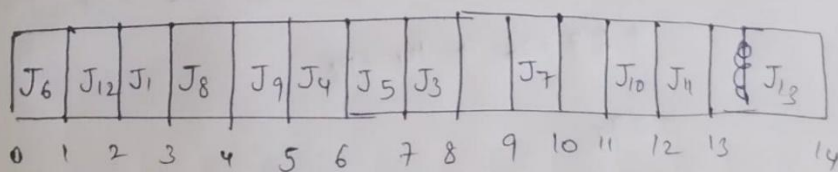
J_2 Reject - J_2



12) J_{13} , $P_{13} = 19$, $D_{13} = 14$



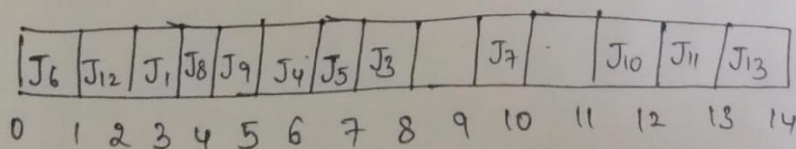
13) J_{11} , $P_{11} = 14$, $D_{11} = 13$



14) J_{14} , $P_{14} = 11$, $D_{14} = 1$

As deadline is 1, there are slots left

for P_{14} so Reject J_{14}



Final Job Sequence,

$$= \{ J_5, J_3, J_4, J_7, J_{12}, J_{10}, J_8, J_9, J_1, J_6, J_{13}, J_{11} \}$$

J ₀	J ₁₂	J ₁	J ₈	J ₉	J ₄	J ₅	J ₃		J ₇		J ₁₀	J ₁₁	J ₁₃	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

Total profit = 21 + 27 + 22 + 25 + 24 + 28 + 30 + 29
+ 27 + 26 + 14 + 19
= 292

CODE:

```
GNU nano 7.2
#include <stdio.h>

#define MAX 20

// Structure for Job
struct Job {
    int id;
    int profit;
    int deadline;
};

// Sort jobs by profit (descending)
void sortJobs(struct Job jobs[], int n) {
    struct Job temp;
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (jobs[i].profit < jobs[j].profit) {
                temp = jobs[i];
                jobs[i] = jobs[j];
                jobs[j] = temp;
            }
        }
    }
}

int main() {
    int n;
    struct Job jobs[MAX];

    printf("Enter number of jobs: ");
    scanf("%d", &n);

    // Input profits and deadlines
    for (int i = 0; i < n; i++) {
```

GNU nano 7.2

```
int n;
struct Job jobs[MAX];

printf("Enter number of jobs: ");
scanf("%d", &n);

// Input profits and deadlines
for (int i = 0; i < n; i++) {
    jobs[i].id = i + 1;
    printf("Enter profit and deadline for Job ");
    scanf("%d %d", &jobs[i].profit, &jobs[i].deadline);
}

// Sort by profit
sortJobs(jobs, n);

// Find maximum deadline
int maxDeadline = 0;
for (int i = 0; i < n; i++) {
    if (jobs[i].deadline > maxDeadline)
        maxDeadline = jobs[i].deadline;
}

int slot[MAX] = {0};
int totalProfit = 0;

// Job sequencing
for (int i = 0; i < n; i++) {
    for (int j = jobs[i].deadline - 1; j >= 0; j--) {
        if (slot[j] == 0) {
            slot[j] = jobs[i].id;
            totalProfit += jobs[i].profit;
            break;
        }
    }
}
```

```
        totalProfit += jobs[i].profit;
        break;
    }
}

// Output result
printf("\nSelected job sequence:\n");
for (int i = 0; i < maxDeadline; i++) {
    if (slot[i] != 0)
        printf("Job %d ", slot[i]);
}

printf("\nTotal Profit = %d\n", totalProfit);

return 0;
}
```

OUTPUT:

```
achyuth@achyuth-VirtualBox:~$ nano job_seq.c
achyuth@achyuth-VirtualBox:~$ gcc job_seq.c -o job_seq
achyuth@achyuth-VirtualBox:~$ ./job_seq
Enter number of jobs: 5
Enter profit and deadline for Job 1: 100 2
Enter profit and deadline for Job 2: 19 1
Enter profit and deadline for Job 3: 27 2
Enter profit and deadline for Job 4: 25 1
Enter profit and deadline for Job 5: 15 3

Selected job sequence:
Job 3 Job 1 Job 5
Total Profit = 142
achyuth@achyuth-VirtualBox:~$
```